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Neuromuscular responses of the hamstring and lumbopelvic muscles during unanticipated trunk perturbations

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ABSTRACT

Hamstring strain often occurs when an opponent unanticipatedly perturbs an athlete's movements. We examined the neuromuscular responses of hamstring and trunk muscles during unanticipated trunk perturbations in athletes with and without a history of hamstring strain injury. Male college athletes (11 with a history of a unilateral hamstring injury and 10 without prior injury) knelt while wearing a chest harness attached to a cable that was pulled backward. They were instructed to resist the force isometrically and maintain their position when the perturbations were applied. The pressure was released with or without a cue (CUE or NoCUE). We measured trunk acceleration, three-dimensional kinematic data, and surface electromyography (EMG) signals of the erector spinae, internal oblique, gluteus maximus, biceps femoris long head, and semitendinosus muscles. Maximum trunk acceleration and displacement were greater with NoCUE in both groups ($p < 0.05$). EMG amplitude did not differ after perturbation of any investigated muscle. The injured group demonstrated a delayed onset of the gluteus maximus and erector spinae muscles in NoCUE versus CUE stimuli ($p < 0.05$). Athletes with a history of hamstring strain injury exhibited a reduced neuromuscular coordination of the lumbopelvic muscles in response to unanticipated trunk movement.

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KEYWORDS

Trunk; pelvis; perturbation; hamstring; lumbopelvic muscles

Introduction

Hamstring muscle strain is the most prevalent injury in team sports such as rugby and soccer, accounting for 12–16% of all injuries (Brooks et al., 2006; Ekstrand et al., 2011; Woods et al., 2004). Most hamstring injuries affect the lateral (i.e., biceps femoris long head) rather than medial hamstring muscles and occur during high-speed sprinting (Arnason et al., 1996; Brooks et al., 2006; Woods et al., 2004). The late-swing phase of the sprint cycle is a potentially injurious moment owing to a more significant strain in the biceps femoris long head than in the rest of the hamstring muscles (Schache et al., 2012). The greater elongation peak of the BFlh (compared with the semitendinosus and semimembranosus muscles) that coincides with the second peak of the anterior pelvic tilt during late swing is influenced by activities generated by the pelvic and trunk muscles (Chumanov et al., 2007; Nagano et al., 2014). This association was most significant during the swing phase and at maximal running speed (Chumanov et al., 2007). Compared with the upright trunk position, the forward trunk leans during sprinting is associated with an increased anterior tilt of the pelvis and hip flexion, resulting in a significantly greater hamstring elongation (Higashihara et al., 2015). Peak trunk flexion angle and hamstring muscle forces upon landing are tightly correlated (Kulas et al., 2010), indicating that increasing trunk

angle flexion increases the hamstring force and moment arm. Therefore, there appears to be an anatomical and functional relationship between the trunk, pelvis, and hamstring units. Indeed, a recent prospective study of soccer players showed that lower levels of anterior pelvic tilt and lateral trunk flexion with a higher activity of the lumbopelvic muscles (gluteus maximus, internal and external abdominal obliques, and erector spinae) during the swing phase of sprinting were associated with a lower risk of injury during follow-up (Schuermans, Danneels et al., 2017). Those authors have predominantly focused on the anatomic and structural association between the trunk, pelvis, and hamstring units, in which adequate proximal control is proposed to be essential for preventing the occurrence of excessive strain within this biarticular muscle unit.

In ball games, such as American football, rugby, or soccer, high-speed sprinting actions and cutting movement patterns are frequently perturbed by opponents in an unanticipated manner. Unanticipated movements may lead to uncontrolled trunk and pelvis displacement during athletic movement, placing the hamstring muscles in a high-strain position and causing injury. Activities of muscles in the lumbopelvic region, including the gluteus maximus, erector spinae, and internal and external oblique muscles, decrease

the hamstring workload during sprinting (Chumanov et al., 2007). In response to rapid voluntary trunk flexion and consequent pelvic tilt increase, hamstring activation is essential to stabilise the pelvis and lumbopelvic muscles. If we consider the neuromuscular inhibition of voluntary hamstring activation following hamstring strain injury (Opar et al., 2013), the response to a sudden unexpected trunk perturbation, common in ball game situations, may lead to the uncontrolled trunk and pelvis movements that would increase hamstring elongation and probably injury risk. An adequate response to sudden trunk loading depends not only on a sufficient muscle force but also on correct muscle recruitment and timing patterns to assure mechanical stability. In this line, previous studies of low back pain found that individuals with a recent history of low back pain exhibit longer response latencies in the trunk muscle than do healthy controls when confronted with sudden loading (Cholewicki et al., 2002; Radebold et al., 2000). The delayed muscle response significantly increases the future risk of low back pain (Cholewicki et al., 2005). Considering these findings, athletes with a history of hamstring strain injury could demonstrate impaired muscle response coordination in the trunk, pelvis, and lower extremity kinetic chain during sudden and unexpected movements, which could be one of the possible mechanisms of risk reinjury. Therefore, it is essential to investigate if the athletes with a history of hamstring strain injury demonstrate the activation deficits or alterations in the trunk-pelvic muscles, with the hamstring muscles, during sudden and unexpected movements.

Therefore, this study aimed at investigating the neuromuscular responses of the hamstring and trunk muscles during unanticipated trunk perturbations in athletes with and without a history of hamstring strain injury. We hypothesised that the hamstring and trunk muscle activities and latencies would increase when the trunk perturbations were not anticipated. Furthermore, athletes with a history of hamstring strain injury

demonstrate altered activation coordination in the muscles stabilising the trunk and pelvis, compared with non-injured athletes.

Materials and methods

Participants

A total of 21 male college athletes participated in this study. Eleven athletes (19.9 ± 1.1 years; 172.2 ± 5.9 cm; 68.6 ± 6.2 kg) had at least one unilateral hamstring strain injury (injured group) (Table 1). Another 10 athletes (18.8 ± 1.5 years; 176.8 ± 4.9 cm; 70.5 ± 5.6 kg) had no history of hamstring strain injury (uninjured group). Exclusion criteria included any athlete with a history of bilateral hamstring strain injury and any athlete who had sustained an anterior cruciate ligament rupture previously or who had sustained an injury to the low back, quadriceps, calf, or groin/hip. A hamstring strain injury was defined as acute pain in the hamstring muscle region sustained during training or competition that prevents the player from participating in training or competition for at least one entire week (Hagglund et al., 2005). All participants had returned to practice and participated in the competition. None of them reported any pain or discomfort in the hamstring region during the experimental protocol. In the injured group, the period after injury was 6–54 months, and the time between the injury and returning to sports was 2–12 weeks. This study was approved by the local human research ethics committee and complied with their requirements for human studies. This study conformed to the Declaration of Helsinki. Written informed consent was obtained from each participant.

Experimental protocol

Before the experiment, the participants performed two bouts of 5 s maximum isometric contraction (MVIC) in the prone leg curl with knee flexion angles set at 30°, hip extension, and back extension and flexion to collect the maximum EMG data for

Table 1. Hamstring strain injury in the injured group (n = 11).

Participant	Age (years)	Affected limb	Location of injury	Time since injury (months)	Rehabilitation duration for most recent injury (weeks)	Total number of hamstring strain injuries sustained	Activity at time of injury
1	20	ND	Proximal SM	7	2	1	Kicking
2	21	ND	Middle BFIh	38	4	1	Sprinting
3	19	ND	Middle BFIh	26	12	1	Sprinting
4	19	ND	Proximal BFIh	27	4	1	Sprinting
5	20	ND	Middle BFIh	9	6	1	Sprinting
6	18	ND	Proximal SM	6	1	3	Stretching
7	20	D	Middle BFIh	18	4	1	Sprinting
8	21	D	Proximal SM	54	2	1	Kicking
9	22	ND	Middle BFIh	10	4	2	Sprinting
10	21	D	Proximal SM	50	2	1	Sprinting
11	22	ND	Proximal BFIh	12	2	1	Sprinting

BFIh, long head of the biceps femoris; SM, semimembranosus; D, dominant; ND, nondominant.

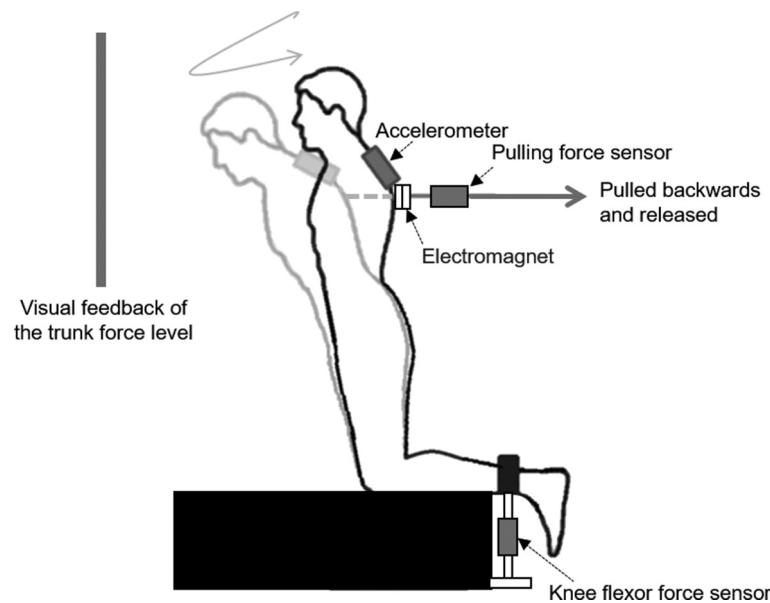


Figure 1. Experimental setup for trunk perturbations. The pulling force was released with anticipated (CUE) or unanticipated (NoCUE) conditions after the resisting force reached the stipulated level.

calculating the normalised EMG activity. Then, the participants knelt on a cushioned board with their ankles secured by a strap and wore a chest harness attached to a cable at the spinous processes of the fifth thoracic vertebrae that was pulled backwards as a resisting force at a constant level corresponding to 18% (12.3 ± 1.1 kgf for the injured group, 12.7 ± 1.0 kgf for the uninjured group) to 20% (13.7 ± 1.2 for the injured group, 14.1 ± 1.1 for the uninjured group) of their weight (Figure 1). A prior empirical study determined the resisting force level, as lower forces would not produce appreciable perturbation and higher forces become uncomfortable when released. The force level was displayed on a screen to provide visual feedback to the participants, helping them reach and maintain the target force. The participants were instructed to isometrically resist the force in the forward direction with the knee angle set at 105° while keeping their arms crossed across their chest and maintaining their initial position as much as possible. Perturbations were applied as a quick force release with an electromagnet in the following conditions: (1) the time of the force release could be anticipated (CUE) and (2) the time of the force release could not be anticipated (NoCUE). When the force release perturbation was applied, the trunk leaned forwards for an instant, and the participants contracted their back muscles and hamstrings to keep the initial trunk position. The perturbation time was indicated to the participants using a countdown tone during the CUE condition. The countdown was a brief tone that counted 3 s before the perturbation, such that the participants would be ready for the perturbations but would not use preparatory actions before the perturbation occurred. Additionally, to examine if the participants contracted their hamstring, gluteus maximus, and erector spinae muscles before the perturbation when the perturbation could be anticipated or if the anticipation only affected the reactive responses, the five-catch trials in which the countdown was made but not perturbed were also randomly incorporated.

Before the experimental data were recorded, participants were allowed to become familiarised with the experimental procedures and the task of reaching and maintaining the target force. Then, they were perturbed once in each condition, and these data were not used in the analysis. The conditions were randomised, and five trials (each with a duration of 30 s) were collected in each condition. Because some participants showed irregular responses before the perturbation was applied in the CUE condition, some trials were rejected when the muscle onset time was identified as an outlier, and it was impossible to determine the onset of muscle activity due to excessive coactivation of muscles. Therefore, we used only the five successful perturbations for qualitative analysis for the prior knowledge condition.

Data acquisition

Trunk acceleration

Linear acceleration of the trunk was measured using a lightweight (35 g) tri-axial accelerometer (LP-WS0902, Logical Product, Fukuoka, Japan). The accelerometer was secured to the spinous processes of the participant's first and second thoracic vertebrae using double-sided sticking plaster. The three axes of the accelerometer were aligned close to the anatomical axes, i.e., the x-axis was aligned mediolaterally, the y-axis vertically, and the z-axis anteroposteriorly. Acceleration data were sampled at a frequency of 200 Hz and were saved in the accelerometer's built-in memory (32 MB). After testing, raw data were downloaded from the accelerometer into a database for analysis.

Knee flexion force

The load cell (MT-150, Sakai Medical, Japan) was used to determine knee flexor force during the trunk perturbation. Participants knelt on a cushioned board, with the ankles

secured by individual ankle straps attached to uniaxial load cells. The knee flexion force was measured from both the left and right limbs.

Three-dimensional kinematics

Joint kinematic data were collected at 200 Hz using an eight-infrared camera Oqus motion analysis system (Qualisys Medical AB, Gothenburg, Sweden). Passive retro-reflective markers were placed on the following landmarks: anterior superior iliac spines, posterior superior iliac spines, and greater trochanters; four cluster markers on the thigh, medial, and lateral femoral condyles; and four cluster markers on the shank, medial and lateral malleoli, and the calcaneal tuber. A neutral stance trial was recorded for each participant before the perturbation tasks and used as reference data for subsequent trials.

Surface electromyography (EMG)

The surface EMG of the biceps femoris long head, semitendinosus, gluteus maximus, erector spinae, and internal oblique muscles (affected limb in the injured group, dominant limb in the uninjured group) were recorded using a surface EMG system (Bagnoli-8, DELSYS Inc., USA). Bipolar surface Ag/AgCl electrodes were placed on the following locations, according to recommendations by SENIAM (Hermens et al., 2000) with an interelectrode distance of 20 mm: the biceps femoris muscle long head, the midpoint of a line connecting the ischial tuberosity and the lateral epicondyle of the tibia; the medial hamstrings, the midpoint of a line connecting the ischial tuberosity and medial epicondyle of the tibia; the gluteus maximus muscle, the midpoint of a line connecting the sacral vertebrae and the greater trochanter; and the erector spinae muscle, two finger-widths lateral from the spinous process of L1. The electrodes for measuring the internal oblique muscle were positioned 2 cm from the medial side of the anterior superior iliac spine (Narouei et al., 2018). Additionally, the surface electrode positions were determined by palpation of each muscle belly during isometric contraction. A reference electrode was placed on the spinous process of C7. The sampling frequency was 2 kHz. The area of each electrode site was shaved using a disposable razor and cleaned with rubbing alcohol-soaked cotton wool. Electrode cables were fixed with elastic tape. This fixing of the electrode cables minimised motion artefacts. All of the recorded analog data were stored on a personal computer using a 16-bit A/D converter (Power Lab.; AD Instruments, Nagoya, Japan) and data acquisition system (Chart 7.0 for Windows; AD Instruments, Nagoya, Japan). The EMG data were synchronised with the resisting trunk force, knee flexion force, trunk acceleration, and kinematic data using the trigger signal.

Data processing

Typical kinematic and muscle activation responses to the perturbation are shown in Figure 2. The moment of perturbation was detected as the trigger at which the electromagnet was released. Trunk acceleration demonstrated a forward and backward peak following the backward direction's first peak due to the force release (Figure 2). Therefore, we calculated amplitude between the first peak in the forward direction and the

maximum value to the backward direction as maximum trunk acceleration during the postural control. Peak knee flexion force of both limbs after the perturbation was also calculated.

We calculated trunk forward lean and hip and knee flexion angles using Cardan angle sequences in Visual3D software (C-motion, Rockville, MD, USA). We also calculated the pelvic anterior tilt angle, which was defined as the pelvic coordinate system's rotation angle around the x-axis in the spatial coordinate system. Maximal displacement after perturbation onset of the trunk forward lean was calculated. The maximum displacement at the moment of the peak trunk flexion was calculated for the pelvis anterior tilt and hip and knee flexion angles.

The EMG signals were rectified and filtered with a low pass at 50 Hz using a sixth-order, zero-lag Butterworth filter. Analysis of the EMG signals included computing (1) background muscle activity, which was defined as the root mean square of the EMG signal in the 50 ms window before the perturbation; (2) the onset time of the muscle response, which was quantified relative to the onset of the trunk perturbation with the integrated profile method, an automatic method described and validated in the previous study (Allison, 2003); and (3) phasic muscle activity, which was defined as the root mean square of EMG signal in the 200 ms window after the muscle onset was normalised as a percentage of the maximum isometric values (% MVIC) of each muscle.

Statistics

All statistical analyses and assumption testing were conducted using IBM SPSS Statistics V25 (IBM Corp). Comparisons between experimental conditions and groups for the maximum trunk acceleration, peak knee flexion force, and kinematic parameters were conducted using two-way repeated-measures analysis of variance (ANOVA). The catch trials were compared with the CUE conditions using the paired *t*-tests to ensure that hamstring, gluteus maximus, and erector spinae muscles were not pre-loaded before the perturbations. Cohen's *d* was used to calculate effect size (ES). Once this was confirmed, only the main experimental conditions were compared, as there was no phasic activity during the catch trials. The two-way ANOVA was conducted in each muscle to determine the differences of EMG activation onset and phasic muscle activity between experimental conditions and groups. Bonferroni's post hoc analysis was conducted if ANOVA showed statistically significant main or interaction effects. Statistical significance was set at $p < 0.05$. Eta squared (η^2) was used measures of effect size in ANOVA.

Results

Trunk acceleration and kinematic parameters

The mean and standard deviation of the maximum trunk acceleration and each kinematic parameter are shown in Table 2. Two-way ANOVA indicated statistically significant main effects of the experimental condition in the maximum trunk acceleration ($F = 7.709$, $\eta^2 = 0.339$, $p = 0.014$). The maximum trunk acceleration was significantly greater during the NoCUE condition than during the CUE condition ($p = 0.014$). In the maximum displacement of the trunk forward lean angle, a statistically

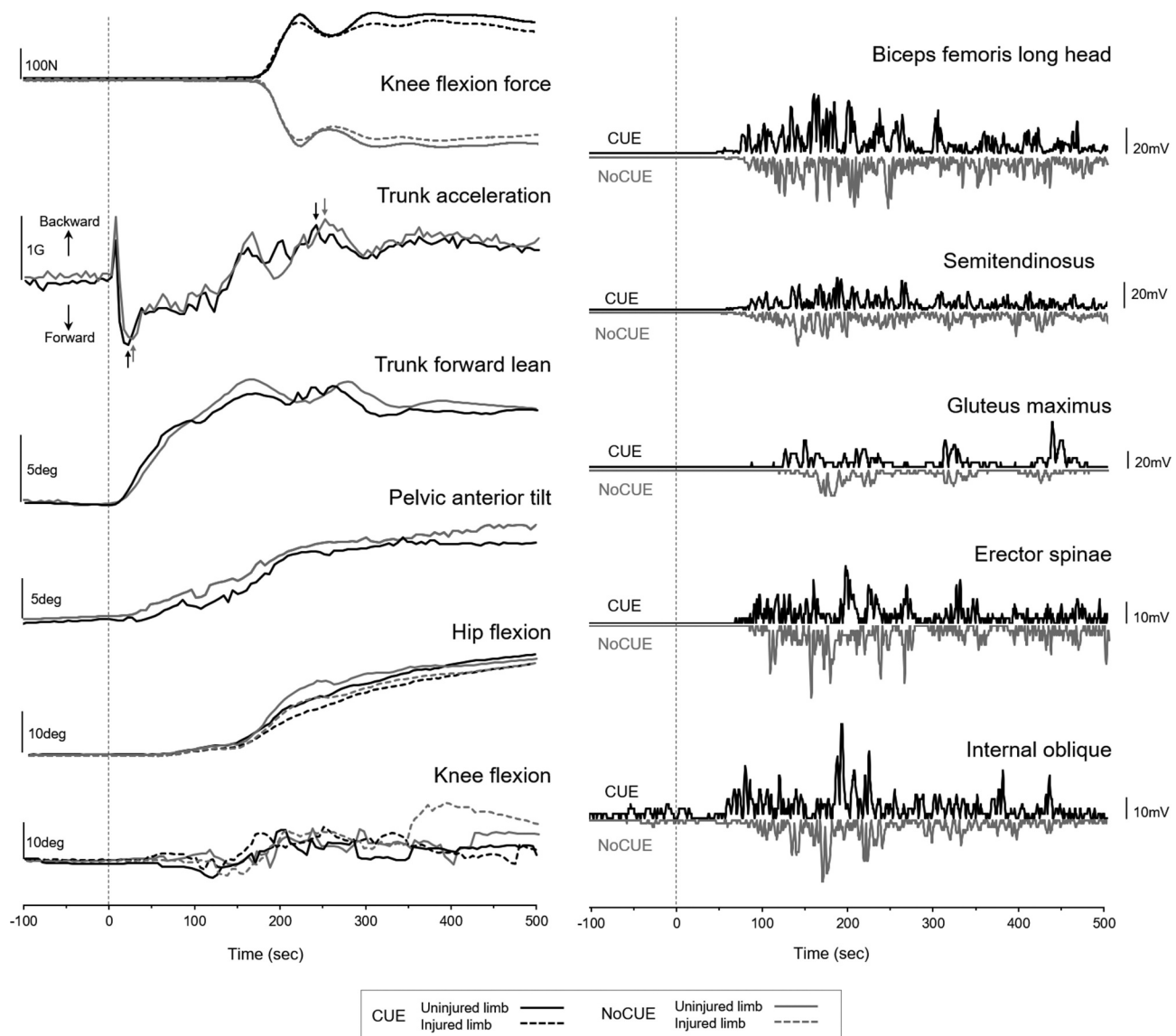


Figure 2. Typical kinematic and muscle activation responses of athletes recruited to the Injured group. Perturbations were applied at time zero with anticipated (CUE, black line) or anticipated (NoCUE, grey line). The solid and dashed lines represent the uninjured limb and injured limb, respectively. For the trunk acceleration, each vertical arrow shows the first peak to the forward direction or maximum value to the backward direction. Amplitude between two peaks was calculated as the maximum trunk acceleration during the postural control.

significant main effect of the experimental condition was observed ($F = 30.837$, $\eta^2 = 0.619$, $p < 0.001$). The post hoc test indicated that the maximum displacement of the trunk forward lean angle was significantly increased during the NoCUE condition than during the CUE condition ($p < 0.001$). There were no differences in the peak knee flexion force, time to peak trunk forward lean, and angular displacement of the pelvis, hip, and knee between experimental conditions in both groups.

EMG activity

There were no differences in the background muscle activity between the CUE condition and catch trials for the biceps femoris long head, semitendinosus, gluteus maximus, and erector spinae muscles in both groups (Table 3). Thus, the muscles were not pre-loaded differently between experimental

conditions. There were no differences in the phasic muscle activity in the 200 ms window between experimental conditions in both groups (Figure 3). The mean and standard deviation of the EMG onset latency in each muscle are shown in Table 4. Details of the statistical results in each muscle are as follows.

Biceps femoris long head and semitendinosus

Two-way ANOVA indicated statistically significant main effects in the onset latency of the biceps femoris long head muscle ($F = 19.996$, $\eta^2 = 0.513$, $p < 0.001$, for experimental condition; $F = 13.511$, $\eta^2 = 0.416$, $p = 0.002$, for group). The post hoc test indicated that the onset latency of the biceps femoris long head was significantly increased during the NoCUE condition than during the CUE condition ($p < 0.001$). The onset latency of the biceps femoris muscle in the uninjured group was

Table 2. Maximum trunk acceleration and kinematic parameters (mean $\pm$ SD).

	Injured (n = 11)		Uninjured (n = 10)	
	CUE	NoCUE	CUE	NoCUE
Maximum trunk acceleration (G) [†]	1.74 $\pm$ 0.21	1.91 $\pm$ 0.26	1.81 $\pm$ 0.16	1.92 $\pm$ 0.20
Peak knee flexion force (N/kg)				
Affected/dominant limb	3.4 $\pm$ 0.7	3.6 $\pm$ 1.0	3.4 $\pm$ 0.5	3.5 $\pm$ 0.3
Unaffected/nondominant limb	3.3 $\pm$ 0.5	3.4 $\pm$ 0.5	3.7 $\pm$ 0.4	3.9 $\pm$ 0.4
Time to peak trunk forward lean (ms)	143.6 $\pm$ 23.1	157.4 $\pm$ 27.0	142.0 $\pm$ 28.2	139.0 $\pm$ 24.0
Angular displacement (deg)				
Maximum trunk forward lean [†]	9.2 $\pm$ 1.9	10.6 $\pm$ 2.6	7.8 $\pm$ 1.3	11.2 $\pm$ 2.7
Pelvis anterior tilt at peak trunk flexion	7.9 $\pm$ 1.8	8.3 $\pm$ 1.8	7.7 $\pm$ 2.0	8.4 $\pm$ 2.4
Hip flexion at peak trunk flexion				
Affected/dominant limb	14.8 $\pm$ 3.2	15.7 $\pm$ 3.5	11.7 $\pm$ 3.7	12.7 $\pm$ 4.1
Unaffected/nondominant limb	15.0 $\pm$ 3.1	15.6 $\pm$ 3.4	11.5 $\pm$ 1.9	11.9 $\pm$ 5.6
Knee flexion at peak trunk flexion				
Affected/dominant limb	19.4 $\pm$ 10.5	20.3 $\pm$ 10.2	14.7 $\pm$ 6.5	15.4 $\pm$ 6.7
Unaffected/nondominant limb	17.8 $\pm$ 10.6	18.6 $\pm$ 12.2	14.4 $\pm$ 5.5	14.3 $\pm$ 5.8

Two-way repeated-measures analysis of variance (ANOVA) was used to compare results between conditions and groups. [†], significant main effect of the experimental condition (CUE/NoCUE). Affected/unaffected applies to the injured group, while dominant/nondominant applies to the uninjured group.

significantly increased compared with that of the injured group ($p = 0.002$). No significant difference was found in the semitendinosus muscle onset latency between experimental conditions or groups.

Gluteus maximus and erector spinae

Significant interaction effects in the gluteus maximus ($F = 8.284$, $\eta^2 = 0.304$, $p = 0.010$) and the erector spinae ($F = 7.007$, $\eta^2 = 0.269$, $p = 0.016$) muscles were observed. The post hoc test showed no differences in muscle latency between experimental conditions in the uninjured group. In contrast, the injured group showed a significant delay of the muscle activation onset during the NoCUE condition ($p = 0.028$ in gluteus

maximus, $p = 0.003$ in erector spinae, Figure 4). Additionally, the gluteus maximus onset latency during the NoCUE condition was significantly increased in the injured group versus the uninjured group ($p < 0.001$, Figure 4). The latency of the erector spinae muscle showed a similar trend: onset latency during the NoCUE condition in the injured group increased compared to the uninjured group ($p = 0.086$).

Internal oblique

In the internal oblique muscle, a statistically significant main effect of the experimental condition was observed ($F = 25.470$, $\eta^2 = 0.573$, $p < 0.001$). The post hoc test indicated that the onset latency of the internal oblique muscle was significantly increased during the NoCUE condition than during the CUE condition ($p < 0.001$, Table 4).

Discussion

This study's primary purpose was to compare the neuromuscular responses of the hamstring and lumbopelvic muscles during anticipated and unanticipated trunk perturbations between the athletes with and without a history of hamstring strain injury. This study demonstrated that the maximum trunk acceleration and displacement concomitantly with a general (except semitendinosus) increase in the involved muscle onset latency were significantly greater during the NoCUE stimulus in both groups. These results supported our hypothesis that muscle onset latency would increase when the trunk perturbations could not be anticipated. We drafted this hypothesis based on previous observations. The trunk quick-release method derived from the study of the reflex muscular response (involuntary) has been extensively used to estimate spine stability via the calculation of trunk stiffness from trunk kinematic response after the force release in the low back pain literature (Cholewicki et al., 2002; Radebold et al., 2000). Several studies have shown that increased muscle activation latency decreases trunk stabilisation (Andersen et al., 2004; Essendrop et al., 2002; Gardner-Morse & Stokes, 2001) and increases subsequent trunk displacement and acceleration (Essendrop et al., 2002; Krajcarski et al., 1999; Stokes et al., 2000) in response to a rapid, unexpected perturbation compared with an expected one (Andersen et al., 2004; Granata et al., 2004; Krajcarski et al., 1999; Stokes et al., 2000). Interestingly, it is also possible to observe a different response within hamstring musculature between CUE and NoCUE conditions when breaking the generated trunk flexion. The increased onset latency of the biceps femoris long head muscle detected along with the rest of the

Table 3. Background electromyography activity for each muscle (mean $\pm$ SD). There were no differences in the background muscle activity between the CUE condition and catch trials. Thus, the muscles were not differently pre-loaded between experimental conditions.

	Injured (n = 11)				Uninjured (n = 10)			
	CUE	Catch	<i>p</i>	ES	CUE	Catch	<i>p</i>	ES
Background activity (μ V)								
Biceps femoris	0.7 $\pm$ 0.9	0.7 $\pm$ 1.2	0.857	0.09	0.5 $\pm$ 1.1	0.4 $\pm$ 0.2	0.387	0.65
Semitendinosus	1.6 $\pm$ 1.9	1.5 $\pm$ 0.8	0.576	0.73	0.8 $\pm$ 0.9	1.0 $\pm$ 0.7	0.113	0.55
Gluteus maximus	0.7 $\pm$ 1.2	0.6 $\pm$ 0.7	0.163	0.40	0.2 $\pm$ 0.5	0.3 $\pm$ 0.4	0.704	0.22
Erector spinae	0.8 $\pm$ 1.2	0.7 $\pm$ 1.0	0.251	0.94	1.0 $\pm$ 1.3	0.5 $\pm$ 0.5	0.131	0.51
Internal oblique	13.9 $\pm$ 10.6	12.9 $\pm$ 11.0	0.544	0.09	8.9 $\pm$ 7.1	8.6 $\pm$ 5.2	0.726	0.05

Cohen's *d* was used to calculate effect size (ES).

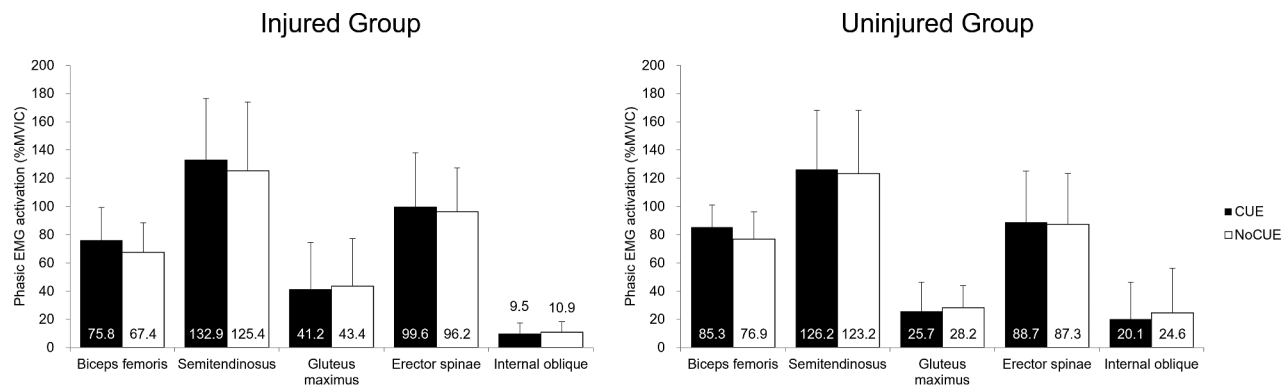


Figure 3. Phasic EMG activation in the 200 ms window after the muscle onset. Data are represented as a percentage of the maximum isometric values (%MVIC) of each muscle. The phasic muscle activation did not differ between experimental conditions and groups.

Table 4. Electromyography onset latency (mean $\pm$ SD).

	Injured (n = 11)		Uninjured (n = 10)	
	CUE	NoCUE	CUE	NoCUE
Onset latency (ms)				
Biceps femoris $\dagger, \#$	61.6 $\pm$ 4.8	70.8 $\pm$ 6.3	67.9 $\pm$ 2.5	72.8 $\pm$ 4.4
Semitendinosus	59.0 $\pm$ 4.3	60.7 $\pm$ 4.6	59.4 $\pm$ 5.7	62.8 $\pm$ 5.2
Gluteus maximus $\#$	119.2 $\pm$ 26.9	145.0 $\pm$ 27.5	113.7 $\pm$ 27.9	103.7 $\pm$ 26.2
Erector spinae $\#$	65.1 $\pm$ 10.6	82.0 $\pm$ 21.8	70.0 $\pm$ 15.6	69.3 $\pm$ 4.2
Internal Oblique $\dagger$	73.5 $\pm$ 15.2	81.0 $\pm$ 10.5	78.8 $\pm$ 13.2	91.2 $\pm$ 13.8

Two-way repeated-measures analysis of variance (ANOVA) with the Bonferroni post hoc test was used to compare results between conditions and groups. $\dagger$, significant main effect of the experimental condition (CUE/NoCUE); $\#$, significant main effect of the group (injured/uninjured); $\#$, significant interaction effect.

lumbo-abdominal musculature during the NoCUE condition compared with the CUE condition was not replicated on the semitendinosus muscle, the latter remaining without any alteration between conditions. This finding may highlight the different roles of the hamstring muscles. A previous study demonstrated that the semitendinosus muscle shows high activation at deeper knee angles (Makiyama et al., 2006) and during eccentric knee flexion exercise due to its morphological ability to effectively handle strain (Ono et al., 2010). Thus, the kneeling position, which was adopted in this study, may require greater activation demand of the semitendinosus than the biceps femoris long head, regardless of conditions of the trunk perturbations.

In the trunk perturbation tasks conducted in this study, the hamstrings were activated first, followed by the erector spinae, and, lastly, the gluteus maximus. Apparent differences in muscle response patterns to sudden trunk loading between persons with a previous hamstring injury and healthy control subjects were observed: the injured group demonstrated significantly delayed onset of the posterior chain muscles with NoCUE. In contrast, no difference was observed in the uninjured group. These results partly supported our hypothesis that the athletes with a history of hamstring strain injury demonstrate altered activation coordination in the muscles stabilising the trunk and pelvis, compared with non-injured athletes. However, there were no differences in knee flexor force between the two groups. The fact that the ability to generate force by the knee flexors did not differ between previously injured and uninjured athletes confirm what has been shown

by other studies (Roe et al., 2020; Van Dyk et al., 2017, 2019). Interestingly, the previously injured subjects did not respond to the delayed onset latency with an increase in trunk displacement or acceleration to the load release, which was previously suggested in low back pain population (Essendrop et al., 2002; Krajcinski et al., 1999; Stokes et al., 2000). In response to a rapid trunk flexion, hamstring activation in coordination with the erectors and gluteus is essential to break the movement (Oddsson & Thorstensson, 1987). Theoretically, a greater anterior tilt of the pelvis would superiorly translate the ischial tuberosity (proximal origin of the hamstring muscles), resulting in a greater active lengthening and passive tension demand of the posterior thigh musculature due to a greater moment arm derived from the relative hip flexion (anterior iliac spine closer to thigh) generated. Delayed response of the posterior chain muscles can induce a greater elongation of the hamstring muscles derived owing to an increase in an anterior pelvic tilt. When athletes were in front of any unexpected disturbance by an opponent during a sprint, cutting, or stopping, they experienced a forced flexion loading/unloading of the upper body, which may lead to increased hamstring muscle strain. The relationships between muscle elongation and strain injury are well understood: unanticipated movements may lead to uncontrolled trunk and pelvis displacement during athletic movement, which may place the hamstring muscles in a high-risk position (high strain) and cause injury. Therefore, conditioning and prevention require evidence-based training to reduce onset latency (Wilder et al., 1996) and increase dynamic trunk stability (Lee & McGill, 2015) to counteract trunk flexion in game situations.

Our results showed that the injured group demonstrated significantly delayed onset of the gluteus maximus and erector spinae muscles during unanticipated trunk perturbation may suggest an alteration in the lumbopelvic neuromuscular coordination following injury. However, no differences were found in phasic muscle activation after perturbation of any investigated muscle. Schuermans et al. (2017) suggested that, during a prone hip extension task, delayed EMG onset rather than EMG amplitude is related to future injury susceptibility due to altered neuromuscular coordination and interplay throughout the hamstring and

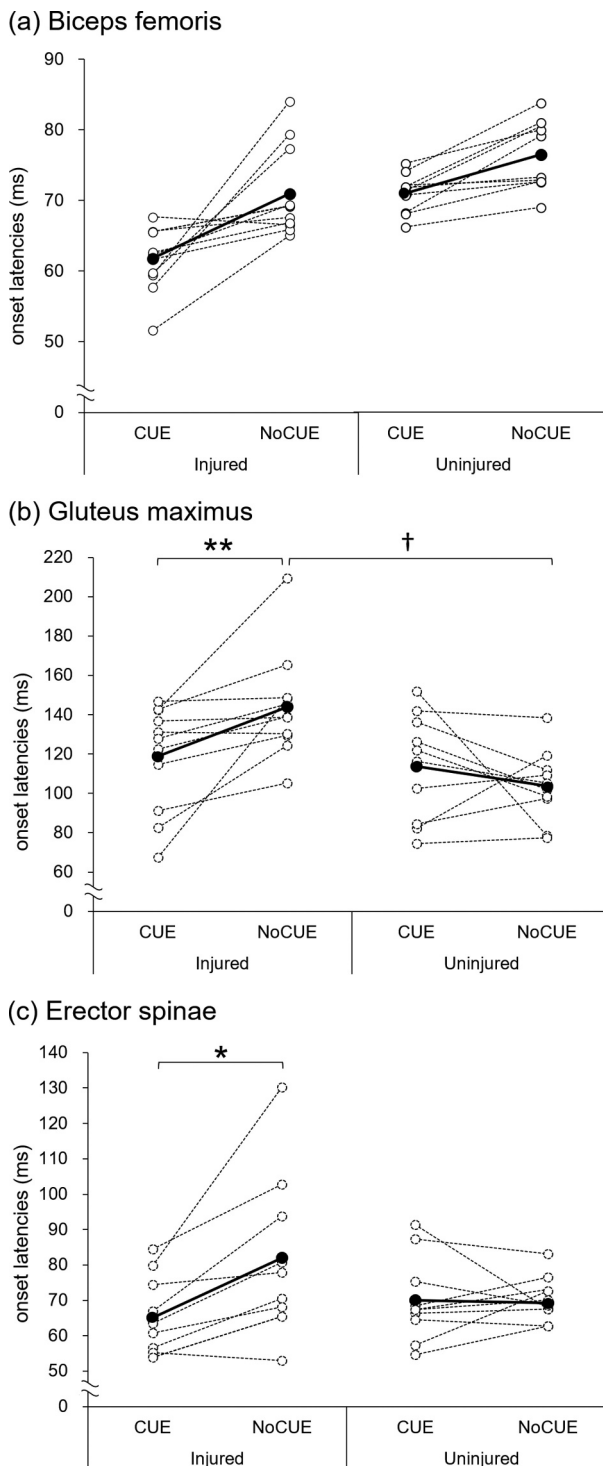


Figure 4. Muscle response latencies in the biceps femoris (a), gluteus maximus (b), and erector spinae (c) muscles. Open circles represent the onset latencies in each subject. Closed circles represent the average value for all subjects. Latencies of the gluteus maximus and erector spinae muscles significantly increased in the NoCUE versus CUE condition in the injured group (* $p < 0.05$, ** $p < 0.01$). In the NoCUE condition, latency for the gluteus maximus was greater in the injured group than the uninjured group († $p < 0.001$).

lumbopelvic muscles (Schuermans, Van Tiggelen et al., 2017). Thus, a significant delay in the onset timing of the lumbopelvic muscles in previously injured athletes may increase the risk of reinjury. Furthermore, our results

demonstrated substantial individual variability of EMG onset between experimental conditions in the injured group: some participants showed a significant delay in onset during the NoCUE condition in the biceps femoris gluteus maximus and erector spinae muscles (Figure 4). These individual variabilities suggest the importance of enhancing the coordination of the hamstring (biceps femoris muscle concretely), gluteus maximus, and erector spinae muscles in rehabilitation practices. This approach can help athletes develop individual-based training approaches to improve lumbopelvic muscle function. Therefore, according to the results of this study and from a hamstring injury rehabilitation perspective, an evidenced motor control training to reduce the onset latency of posterior chain muscles (Park et al., 2016) and anterior pelvic tilt (Mendiguchia et al., 2021; Park et al., 2016) would be convenient to consider within a multifactorial and individualised hamstring rehabilitation approach (Mendiguchia et al., 2021).

This study has some its limitations. First, the period after injury (6–54 months) was large. However, previous studies have suggested that activation deficits (Opar et al., 2013) and insufficient muscle contraction (Nagano et al., 2015) persist despite successful rehabilitation and a return to preinjury levels of training and playing. This neuromuscular inhibition is considered to limit the extent of muscular adaptations in response to rehabilitative and prophylactic exercises by central mechanisms (Opar et al., 2015). Furthermore, another study has reported that an athlete with a previous hamstring strain injury that had occurred more than 5 years ago also demonstrated an activation deficit in the BFLh muscle during sprinting (Higashihara et al., 2019). Considering these studies, in the case of some athletes, the function of the BFLh muscle following hamstring strain injury may not have been entirely recovered despite rehabilitation. Second, the exact cause or effect of significantly delayed onset of the posterior chain muscles during unanticipated trunk perturbation in individuals with previous hamstring injury cannot be properly addressed with the current study.

Conclusively, we investigated the neuromuscular responses of the hamstring and trunk muscles during anticipated and unanticipated trunk perturbations in athletes with and without a history of hamstring strain injury. We found that athletes with a history of injury demonstrated significantly delayed onset of the posterior chain muscles during unanticipated trunk perturbation, indicating a reduction in the neuromuscular coordination of the lumbopelvic muscles. These results provide a basis for improved, evidence-based rehabilitation and prevention, mainly focusing on increasing lumbopelvic neuromuscular coordination following injury.

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Appendix. Relationship between the internal oblique activation before the perturbation and maximum pelvic displacement

The relationship between the pelvis anterior tilt displacement and the internal oblique (IO) muscle activity before the sudden trunk perturbation for all participants was investigated. The maximum displacement of the anterior pelvis tilt was calculated. The root mean square of the internal oblique EMG signal in the 50 ms window before the

perturbation was normalised as a percentage of the maximal voluntary contraction (%MVC). Pearson's correlation test was performed to assess the relationship between the internal oblique activation before the perturbation and maximum displacement of the anterior pelvic tilt. Internal oblique activity before the perturbation was inversely correlated with the maximum anterior pelvic tilt displacement in both experimental conditions (CUE: $r = -0.641$, $P = 0.002$; NoCUE: $r = -0.713$, $P < 0.001$).

